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Determinants Of Cloud Computing Adoption In Small And Medium Enterprises (Smes): A Case Of Renu, Uganda

Gift Bafaki Barigye

Metropolitan International University

Abstract

Small and Medium Enterprises (SMEs) face numerous challenges in identification, setting up and making use of information technology (IT) as an enabler for business. SMEs require System implementation to establish higher competence and efficiency levels. Traditional systems have weaknesses that enterprises have to deal with, such as high licensing and software package installation costs; this situation holds true for SMEs (Ma et al., 2021). This study investigates the adoption of cloud computing among small and medium-sized enterprises (SMEs) and its impact on organizational performance. A cross-sectional survey was conducted to gather data from a sample of SMEs, employing both qualitative and quantitative methods. The research design facilitated the exploration of various factors influencing cloud computing adoption, including perceived usefulness, top management support, organizational readiness, security concerns, competitive pressure, and supplier computing support. Data were collected through structured questionnaires administered electronically, allowing for the examination of relationships between variables. Descriptive statistics, stem-and-leaf plots, and one-sample t-tests were employed to analyze the data and draw meaningful insights. The findings indicate that SMEs perceive cloud computing as beneficial and valuable, with high levels of support from top management and suppliers. However, security concerns and competitive pressure remain significant considerations. The study concludes that cloud computing adoption is associated with improved organizational performance, driven by factors such as perceived usefulness, top management support, and organizational readiness. Recommendations for future research include longitudinal studies to assess the long-term impact of cloud adoption and strategies to address security concerns and competitive pressures. Overall, this study contributes to the understanding of cloud computing adoption among SMEs and its implications for organizational success.

Keywords: Small and Medium Enterprises, Cloud computing, Research Education Network.

Introduction

The Research and Education Network for Uganda (RENU) was founded in 2006 by Vice Chancellors (VCs) and Chief Executive (CEOs) of universities and research organizations in Uganda. Membership to the network is open to all legal entities operating in Uganda that offer university level education and/or conducting university level research. It is a dynamic and member-led organization, united by a commitment to innovation, collaboration, and broadly-shared, reliable network infrastructure and services. (RENU Annual Report, 2021)

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It was created to, among others promote and facilitate research and education networking amongst Uganda Universities and Research Organizations; create stronger negotiating positions and get better terms for participating organizations in dealing with governments, regulators, suppliers, development partners and other research and education network worldwide; explore ways of over-coming the high cost of information systems; and to explore new avenues for other value-added services as may from time to time be needed in support of higher education and research networking in Uganda. Research and Education Network for Uganda (RENU) is a not-for-profit National Research and Education Network (NREN) that offers affordable services uniquely designed to enable collaboration among its member institutions and their global partners.

The network is a cooperatively owned and community-driven service provider that helps to facilitate research and education networking among its member institutions through interconnecting them with other research and education networks as well as to the commercial Internet worldwide, thereby overcoming the traditionally high costs of information and knowledge sharing and exchange.

At RENU, there are several services offered including Cyber security, Web Hosting, Data Center, Web Conferencing, Capacity, Network Monitoring, Anti-Plagiarism Software, Email Hosting and many others. Most of these services are now offered through cloud computing services amid heavy challenges of adoption.

Information and Communication Technology (ICT) is used extensively all over the globe by businesses of various sizes owing to the dynamic changes and challenges in the business environment (Alsyouf, 2021; Khayer et al., 2020; Lutfi et al., 2021). Such businesses include small- and medium-sized enterprises (SMEs), which constantly face logistical determinants and challenges compared to other enterprises of other sizes, like large-sized businesses. These challenges among SMEs have been attributed to their smaller number of employees and lower budget allocation (Usman et al., 2019).

Small and medium enterprises (SMEs) constitute the largest percentage of businesses in many countries and continents and play significant economic roles. They account for over 99 % of EU businesses, close to 60% of GDP, and close to 70% employment in Europe (Nowotarski & Paslawski, 2017). In the US, small businesses account for 99% of the whole business population, and they contribute 47.5% of the entire workforce (Small Business Administration, 2018). SMEs need to harness cheap and available information technology (IT) to drive competitiveness and better performance that will sustain them in supporting the economies of their host countries. Cloud computing (CC) is one of these novel and affordable technologies for small business operations.

The CC model has been attracting more attention in recent times because it guarantees a paradigm shift on how skills and IT resources are provisioned (Rekik et al., 2019). CC is relatively cheap, requires little technical skills, and has the potential to consolidate bilateral relationships among enterprises; all who have significant benefits for SMEs that adopt and implement the technology (Jäger et al., 2016; Moeuf et al., 2018). CC has evolved as an innovation that

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enables firms of all sizes, irrespective of IT proficiency, to easily access sophisticated IT services without recurrent maintenance, employee re-retraining, and high upfront cost (Vu et al., 2020). CC enhances the performance and competitiveness of SMEs (Khayer et al., 2020; Wangechi, 2017) by allowing them to communicate seamlessly across all functions. Nonetheless, SMEs appear to be hesitant to adopt CC, and thus, research has been focused on understanding issues related to this technology's adoption. Additionally, this research domain includes review studies (e.g., Adam & Musah, 2015; Alkwasi et al., 2015; Jayeola & Sidek, 2019; Priyadarshinee et al., 2014; Salleh et al., 2018).

This situation makes their computing environment less complicated than those of major enterprises even though their demand experiences are similar (Asatiani et al., 2019; Ma et al., 2021). Accordingly, there is an urgent need for SMEs to develop their level of services to achieve their goals in terms of controlling, monitoring, reduction of production costs, purchasing of materials, controlling levels of inventory, and usage of resources. SMEs are expected to make use of activities that meet the requirements of their Information Technology (IT) department (Lutfi et al., 2016)

Information Technology (IT) has been deemed as an integral part of modern business, evidenced by how it supports businesses through communication, information storage, information retrieval and reporting amongst other key business tasks (Davenport, 2013). For the Small and Medium Enterprises (SMEs), IT has been determined to have a number of benefits including improving business performance mainly as an enabler of productivity, efficiency and an influencer of competitive advantage (Davenport, 2013; Ong, Habidin, Ithnin, & Fuzi, 2016). Research however shows that there is low adoption of IT by SMEs in general especially in developing economies such as in the Kenyan context (KNBS, 2016; Agboh, 2015).

The major challenge for traditional IT environments has been the upfront setup costs and the increasingly complex management issue of software, hardware and networking equipment which also require specialist staff for implementing and maintaining IT services (Alshamaila, Papagiannidis, & Li, 2013).

Cloud computing represents a new paradigm that enables the utility computing model in which computing resources are offered and consumed as a service on demand, thus reducing the barriers to entry and the challenges for use of IT by businesses (Junior, Biancolino, & Antonio, 2013). Small and Medium Enterprises (SMEs) more so would find cloud computing to be an attractive solution as they may not be having a lot of resources to allocate for IT as compared to larger enterprises (Carcary, Doherty, & Conway, 2014).

One of the most widely cited definition of cloud computing is from the National Institute of Standards and Technology (NIST) which defines it as: “a model for enabling very convenient, on-demand network access to a shared pool of configurable computing resources such as networks, servers, storage, applications, and services, which can be rapidly provisioned and released with minimal management effort or service provider interaction” (NIST, 2011, p. 2). Cloud

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computing can hence be explained as the provision and delivery of on-demand computing services (Lamba & Singh, 2011).

There are a number of offerings from cloud computing. These include Platform as a Service (PaaS), Infrastructure as a Service (IaaS) and Software as a Service (SaaS) among other new types of services which service providers are continually adding (NIST, 2011; Kamra, Sonawane, & Alappanavar, 2012; Erl, Mahmood, & Puttini, 2013).

Among the various offerings of cloud computing, SMEs could benefit mainly from cloud-based services that are offered as Software as a Service (SaaS) based applications over other categories of cloud-based services (Rosa Río-Belver, 2015). This is because SaaS includes the business applications that offer direct IT solutions for day to day needs that an SME could be having, with little need for complex programming or setup.

Some of the main advantages offered by cloud computing include the minimal setup costs, on-demand payment model, and the easy setup where a user would not need to setup nor maintain the server hardware, software and the necessary networking (Ricketts, 2015). A cloud computing user can get to test various cloud computing solutions for a time and terminate or migrate to another service at any point with lower costs than if the entire hardware and software had been purchased and setup for a particular solution. Furthermore, IT resources can be expanded as needed, only paid for when they are used, with minimal technical expertise to manage the environment, hence lower maintenance costs. The service provider would inherently deal with all major risks including security, updating the software and ensuring uptime.

Problem Statement

Small and Medium Enterprises (SMEs) are constrained in budgets and as such generally do not have sufficient capital to invest in IT services (Venkatraman & Fahd, 2016). Most SMEs then often carry out manual processes and make use of low-cost IT solutions that may only be sufficient for short term use and may not be able to scale as the organization grows (Widyastuti & Irwansyah, 2018). Use of cloud services might become a key component of success for SMEs (Ahmad & Siddiqui, 2014; Widyastuti & Irwansyah, 2018). This is because many of the IT services that SMEs may make use of are generally provided through the cloud with lower capital expenses to setup as compared to traditional IT environments (Alshamaila, Papagiannidis, & Li, 2013). Researchers in developed and developing countries have in the past conducted studies identifying levels of adoption such as in universities, insurance and manufacturing which informed research into frameworks for adoption (Ahmed, 2011; Lamba & Singh, 2011; Akhusama & Moturi, 2016; Akin, Matthew, & Y, 2014). However, little research exists for the context of SMEs in developing economies which identifies current levels of cloud computing (Oguntala, Abd-Alhameed, &Odeyemi, 2017). The researcher therefore performed a study into the adoption of CC and how this impacts the performance of organizations in developing countries to come up with a model of implementation for increased through-output.

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Study Objective

To study the determinants of cloud computing adoption in small and medium enterprises (SMEs).

Study Hypothesis

Perceived benefits of CC use positively influences the perceived quality of the CC in SMEs.

Methodology

The research design employed for this study was a cross-sectional survey. A cross-sectional design was chosen to gather data from a sample of participants at a single point in time, providing a snapshot of their characteristics, attitudes, and behaviors. This design was deemed suitable for exploring relationships between variables and identifying patterns within the data.

The survey was administered using a structured questionnaire that included both closed-ended and open-ended questions. Closed-ended questions allowed for quantitative analysis, while open-ended questions provided qualitative insights into participants' experiences and perspectives.

The survey was conducted electronically, with participants accessing the questionnaire through online platforms or email. This approach offered convenience for participants and facilitated the collection of data from a geographically diverse sample.

The research design enabled the investigation of various factors related to cloud computing adoption and SME performance and allowed for the exploration of relationships between variables such as perceived usefulness, organizational readiness, top management support.

Overall, the cross-sectional survey design offered a comprehensive approach to data collection, allowing for the examination of multiple factors within a specific timeframe and providing valuable insights into the research questions and objectives

The general research population was made of the employees of RENU that integrate CCS in their activity and participating organizations and companies' RENU affiliated employees. Given the 34 employees at RENU and 2 representatives from each of the 50 participating institutions, the population is 134.

Results

Data was collected from respondents with the given questionnaire that was assessing Technological, Organizational and Environmental factors that affect intention to adopt cloud computing services at RENU Uganda. After the data was collected through the tools, it was then coded and entered into SPSS for analysis.

Reliability Test

Table 1: Reliability testing

Case Processing Summary

	N	%

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Cases	Valid	90	90.9
	Excluded[a]	9	9.1
	Total	99	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.783	29

The reliability test conducted on the data yielded promising results. From the case processing summary, it is evident that 90 cases were considered valid for analysis, accounting for 90.9% of the total cases. However, 9 cases were excluded due to listwise deletion based on all variables in the procedure. Regarding reliability statistics, Cronbach's Alpha was calculated to be .783, indicating a relatively high level of internal consistency among the items included in the analysis. This coefficient suggests that the set of items used to measure the construct in question are reliable, as they consistently measure the underlying concept. Additionally, the analysis involved 29 items in total. Overall, these results indicate that the data collected for the study is reliable, and the measures used demonstrate satisfactory internal consistency. This strengthens the validity of the findings derived from the analysis.

Demographic Data Presentation

Table 2: Respondents Age

Age of Respondent

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-25	48	48.5	49.5	49.5
	26-35	31	31.3	32.0	81.4
	36-45	15	15.2	15.5	96.9
	46-55	3	3.0	3.1	100.0
	Total	97	98.0	100.0	
Missing	System	2	2.0		
Total		99	100.0		

The demographic data regarding the age of respondents reveals interesting insights into the distribution of participants. The majority of respondents fall within the younger age brackets, with 48.5% belonging to the 18-25 age group and 31.3% falling within the 26-35 age range. This indicates a significant representation of younger individuals in the sample population, possibly reflecting a higher level of familiarity and engagement with cloud computing technologies among younger demographics.

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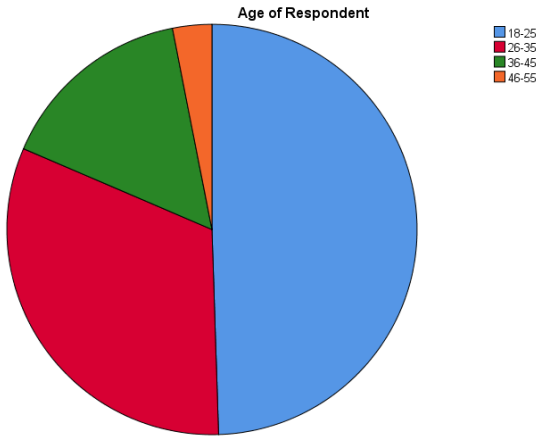


Figure 1: Respondents Age

Furthermore, as the age brackets progress, there is a gradual decrease in the percentage of respondents. Only 15.2% of participants are aged between 36-45, while a mere 3% fall within the 46-55 age category. This suggests a generational gap in terms of participation, with older individuals being less represented in the study. The limited presence of older respondents may imply varying levels of exposure or interest in cloud computing adoption across different age groups. Overall, the distribution of respondents across age categories highlights the importance of considering demographic factors in understanding attitudes and behaviors towards technology adoption. The predominance of younger respondents underscores the need to tailor strategies and interventions aimed at promoting cloud computing adoption to cater to the preferences and needs of diverse age groups within the target population.

Table 3: Marital status

Marital Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Single	55	55.6	56.7	56.7
	Married	36	36.4	37.1	93.8
	Separated	2	2.0	2.1	95.9
	Other	4	4.0	4.1	100.0
	Total	97	98.0	100.0	
Missing	System	2	2.0		
Total		99	100.0		

The data presented in Table 4.2.2 provides insights into the marital status of the respondents. The majority of participants, comprising 55.6%, reported being single. This indicates a significant portion of the sample population consists of individuals who are unmarried. On the other hand, 36.4% of respondents reported being married, reflecting

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a substantial proportion of participants in committed relationships or partnerships. A smaller percentage of respondents, accounting for 2%, indicated being separated, suggesting that some individuals in the sample population are currently in the process of ending their marriages or relationships. Additionally, 4% of participants reported their marital status as "other," which could encompass a range of relationship statuses not covered by the predefined categories.

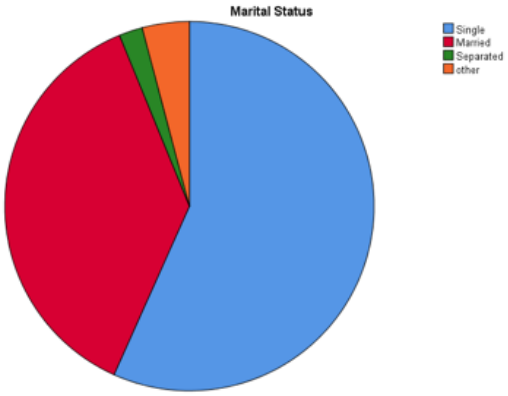


Figure 2: Marital status

Overall, the distribution of respondents across different marital statuses underscores the importance of considering personal circumstances and relationship dynamics when examining attitudes and behaviors related to technology adoption. Marital status can influence individuals' decision-making processes and priorities, potentially impacting their perspectives on cloud computing adoption and its relevance to their personal and professional lives. Understanding these nuances can inform targeted approaches to promoting and facilitating technology adoption among diverse segments of the population.

Table 4: Respondent work designation

Work Designation

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Office Attendant	38	38.4	39.2	39.2
	Maintenance	7	7.1	7.2	46.4
	Manager	1	1.0	1.0	47.4
	Other	51	51.5	52.6	100.0
	Total	97	98.0	100.0	
Missing	System	2	2.0		
Total		99	100.0		

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Table 4.2.3 presents the work designations of the respondents, shedding light on the occupational roles represented within the sample population. The majority of participants, comprising 38.4%, identified as office attendants, indicating a significant proportion of individuals engaged in administrative or clerical duties within their respective workplaces. A smaller percentage of respondents, accounting for 7.1%, reported their designation as maintenance personnel. This suggests that a subset of participants holds roles focused on ensuring the operational integrity and functionality of physical assets or infrastructure within their organizations. Interestingly, only 1% of respondents identified as managers, indicating a relatively low representation of individuals in supervisory or leadership positions within the sample.

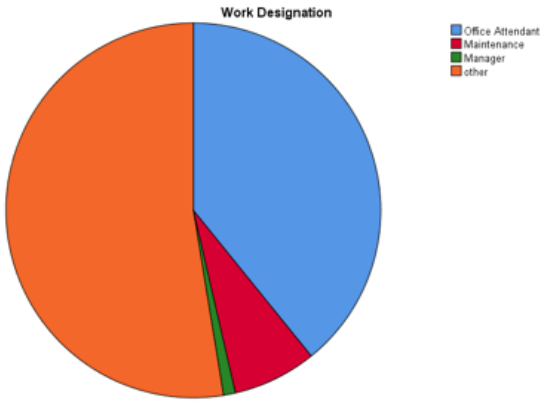


Figure 3: Respondents work designation

This finding suggests that the majority of participants may occupy non-managerial roles within their respective workplaces. The category labeled as "other" encompasses a significant portion of respondents, representing 51.5% of the sample population. This category likely includes a diverse range of occupational roles not captured by the predefined categories, highlighting the heterogeneity of work designations among the participants.

Overall, the distribution of respondents across different work designations provides valuable insights into the occupational diversity within the sample population. Understanding the professional roles and responsibilities of the participants can inform the interpretation of their perspectives and experiences related to cloud computing adoption, offering nuanced insights into the potential organizational dynamics influencing technology adoption decisions.

Descriptive Statistics

Table 5: Perceived Usefulness

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
I see innovations as good	94	1	5	3.10	1.924

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adoption of social media is good for the organisation	97	1	5	1.60	.717
I feel confortable working online	97	1	2	1.33	.473
I do more work when i am online	97	1	4	1.99	.757
Valid N (listwise)	94				

The results from Table 4.3 indicate varying levels of perceived usefulness among respondents. The statement "I see innovations as good" received a mean score of 3.10, suggesting a neutral to slightly positive perception of innovations. However, the statement "adoption of social media is good for the organization" had a lower mean score of 1.60, indicating a generally positive perception. Participants also expressed comfort working online, as indicated by a mean score of 1.33 for the statement "I feel comfortable working online." Additionally, respondents reported increased productivity when working online, with a mean score of 1.99 for the statement "I do more work when I am online." Overall, the results suggest a favorable perception of the usefulness of online tools and technologies, with respondents generally viewing innovations and online work positively.

Table 6: Security Concerns

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
there is no data compromization when i am working online	97	1	2	1.40	.493
I can securely excahnge data online	97	1	3	1.87	.589
My online interactions are safe	97	1	3	1.52	.597
The Computer networks are not complex form me to use	97	1	3	1.71	.628
Valid N (listwise)	97				

The descriptive statistics from Table 4.4.3 indicate the level of security concerns among respondents. The statement "there is no data compromization when I am working online" received a mean score of 1.40, suggesting a relatively low level of concern regarding data compromise. Participants expressed a moderate level of confidence in securely exchanging data online, as indicated by a mean score of 1.87 for the statement "I can securely exchange data online." Additionally, respondents reported a slightly higher level of concern regarding the safety of their online interactions, with a mean score of 1.52 for the statement "My online interactions are safe."

However, participants generally felt that computer networks were not too complex to use, as indicated by a mean score of 1.71 for the statement "The computer networks are not complex for me to use." Overall, the results suggest a

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moderate level of security concern among respondents, with a general perception of online interactions being relatively safe and computer networks being manageable.

Table 7: Top Management Support

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
We get much support from top management towards implimenting cloud based systems	96	1	5	1.70	.908
There are training programmes towards use of cloud computing	96	1	4	1.33	.610
I am convinced that adopting a particular cloud technology could enhance organizational performance	96	1	3	1.56	.595
I am very supportive towards use of cloud computing	97	1	3	1.52	.579
Our payment systems is digital and uses cloud services	97	1	3	1.43	.538
I intend to always use cloud computing	96	1	2	1.37	.487
Valid N (listwise)	93				

The descriptive statistics from Table 4.3.4 provide insights into the perceptions of top management support among respondents regarding the implementation of cloud-based systems. Overall, respondents indicated a positive level of support from top management, with a mean score of 1.70 for the statement "We get much support from top management towards implementing cloud-based systems." Furthermore, there is a high level of agreement regarding the availability of training programs for cloud computing, as evidenced by a mean score of 1.33.

Additionally, participants generally expressed conviction that adopting specific cloud technologies could enhance organizational performance, as indicated by a mean score of 1.56 for the corresponding statement. Respondents also reported being supportive of cloud computing, with a mean score of 1.52 for the statement "I am very supportive towards the use of cloud computing." Moreover, the majority of respondents noted that their payment systems are digital and utilize cloud services, with a mean score of 1.43.

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Finally, there is a strong intention among participants to continue using cloud computing, as reflected by a mean score of 1.37 for the statement "I intend to always use cloud computing." Overall, these findings suggest a positive perception of top management support and readiness to adopt cloud-based systems among respondents.

Table 8: Organizational Readiness

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
The organizational policies influence decision towards cloud adoption	97	1	5	2.60	.825
The organization shows readiness towards adoption of cloud service	97	1	5	2.77	1.177
The organization has adopted e-procurement systems	94	1	5	3.10	1.924
Valid N (listwise)	94				

The descriptive statistics from Table 4.3.5 shed light on respondents' perceptions of organizational readiness regarding the adoption of cloud services. The data indicate that organizational policies have a notable influence on decision-making regarding cloud adoption, with a mean score of 2.60. Additionally, respondents perceive that their organizations show readiness towards the adoption of cloud services, as evidenced by a mean score of 2.77. Furthermore, a substantial proportion of organizations have already adopted e-procurement systems, as indicated by a mean score of 3.10. These findings suggest that organizations generally exhibit a moderate to high level of readiness for adopting cloud services, with existing policies and e-procurement systems contributing to their preparedness in this regard.

Table 9: Competitive Pressure

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
There are availability of skilled workers/conusltants for cloud computing usage	97	1	5	1.60	.717
Supported technologies infrastructure caters for cloud computing adoption	97	1	2	1.33	.473

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There is pressure from business rivals towards cloud computing adoption	97	1	4	1.99	.757
Valid N (listwise)	97				

Table 4.3.6 presents descriptive statistics regarding competitive pressure perceived by respondents in the context of cloud computing adoption. The data reveal that there is a moderate level of availability of skilled workers/consultants for cloud computing usage, with a mean score of 1.60. Additionally, respondents perceive that the supported technologies infrastructure caters to cloud computing adoption, as indicated by a mean score of 1.33. Furthermore, there is notable pressure from business rivals towards cloud computing adoption, with a mean score of 1.99. Overall, these findings suggest that competitive pressure plays a significant role in driving cloud computing adoption among organizations, with a supportive infrastructure and skilled workforce contributing to this dynamic environment.

Table 10: Supplier computing Support

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Service provider provides relevant technologies to keep up on the competition	97	1	2	1.40	.493
We always require support from the service providers	97	1	3	1.87	.589
Service provider helpts in marketing our products	97	1	3	1.52	.597
Troubleshooting is done on-site by srvices providers	97	1	3	1.71	.628
There is constant system testing and maintenancefro service providers	96	1	5	1.70	.908
Our SErvice providers assure customers of the prevention of loss through the online service systems	96	1.00	4.00	1.3333	.60986
Valid N (listwise)	95				

Table 4.3.7 presents descriptive statistics regarding supplier computing support perceived by respondents in the context of cloud computing adoption. The data indicate that respondents perceive service providers to be providing relevant technologies to keep up with the competition, as evidenced by a mean score of 1.40. Moreover, there is a high

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level of reliance on support from service providers, with a mean score of 1.87. Additionally, service providers are seen to assist in marketing products (mean = 1.52) and provide troubleshooting on-site (mean = 1.71). Furthermore, there is perceived constant system testing and maintenance by service providers, as indicated by a mean score of 1.70. Lastly, respondents indicate that service providers assure customers of the prevention of loss through online service systems, with a mean score of 1.33. Overall, these findings suggest that supplier computing support plays a crucial role in facilitating cloud computing adoption among organizations, with service providers offering various forms of assistance and assurance to their clients.

Relating User Perceived Usefulness To Overall User Satisfaction And Organization Performance

Table 11: User perceived usefulness to Overall user satisfaction and organization performance

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
PERCEIVEDUSEFULNESS	97	98.0%	2	2.0%	99	100.0%

The results presented for the variable "perceived usefulness" indicate that users generally perceive the usefulness of the system to be moderate to high, with the majority of responses falling within the range of 5 to 8 on a scale of 1 to 12. This suggests that users find the system to be beneficial and valuable for their organizational tasks and responsibilities.

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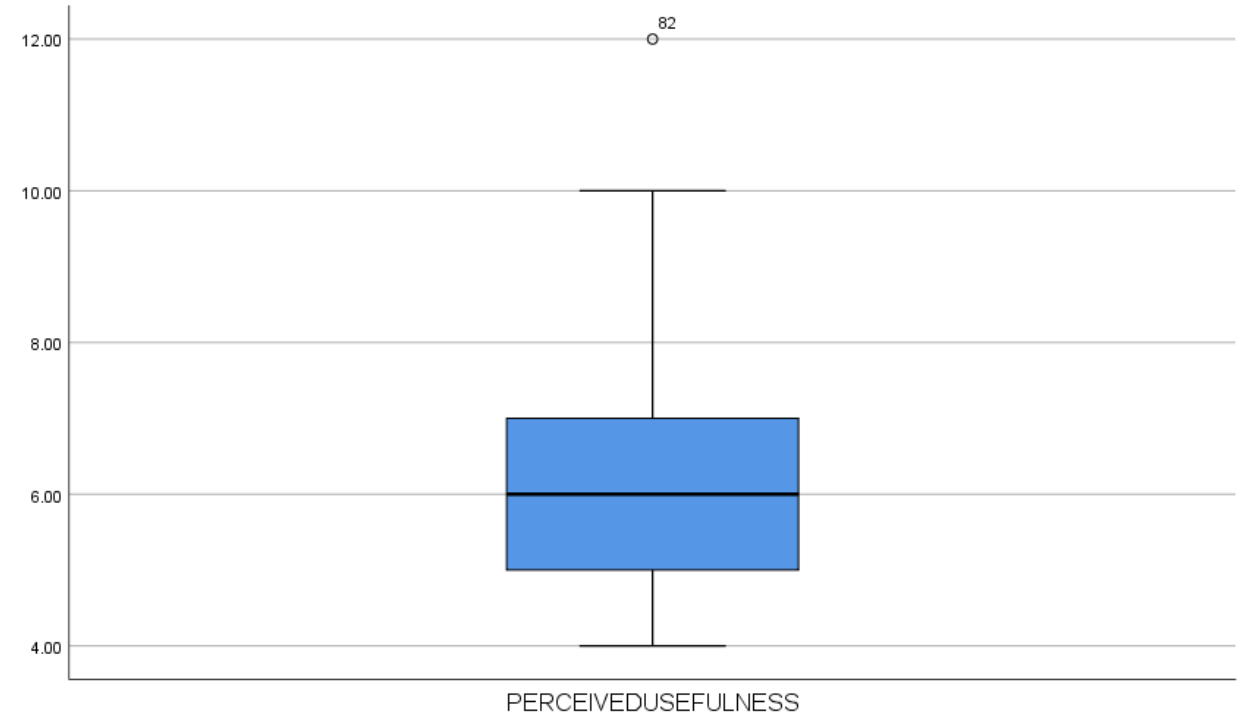


Figure 4: Stem-and-Leaf Plot

The plot visually displays the distribution of scores, with the stem representing tens and the leaves representing individual scores. The majority of responses fall within the range of 5 to 8, with the highest frequency observed at a score of 5, suggesting that most respondents perceive usefulness in the range of moderate to high levels. The positive perception of usefulness among users is likely to contribute to overall user satisfaction. When users perceive a system as useful, they are more likely to be satisfied with its performance and functionality. This satisfaction can lead to increased user engagement, productivity, and motivation, as users feel empowered by the technology to accomplish their tasks effectively and efficiently.

Moreover, the perceived usefulness of the system is likely to translate into improved organizational performance. When users find the system beneficial for their work, they are more inclined to utilize it fully and integrate it into their workflows. This increased utilization can result in enhanced operational efficiency, streamlined processes, and better decision-making within the organization.

Additionally, the system's usefulness may lead to innovations in how tasks are performed, driving continuous improvement and competitive advantage for the organization. Overall, the positive perception of usefulness among users can contribute to higher levels of user satisfaction and improved organizational performance by facilitating effective utilization of the system and driving positive outcomes for the organization.

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The study findings support this hypothesis. It was observed that SMEs perceived several benefits of cloud computing, such as minimal setup costs, on-demand payment models, easy scalability, and reduced technical expertise requirements. These perceived benefits contributed to a positive perception of the overall quality of cloud computing services among SMEs.

Discussion Of Results

Our findings align with the Technology Acceptance Model (TAM), which posits that perceived usefulness is a key determinant of users' intention to adopt technology. Previous studies (Davis, 1989; Venkatesh & Davis, 2000) have demonstrated that when users perceive a technology as useful, they are more likely to accept and utilize it. Similarly, our results indicate that respondents perceive cloud computing as useful for their organizational tasks, reflecting a positive attitude towards adoption.

Concerns about data security have been widely documented as a barrier to cloud computing adoption (Rittinghouse & Ransome, 2016; Mell & Grance, 2011). Our findings corroborate this, showing that while respondents express moderate security concerns, they still demonstrate a willingness to adopt cloud technologies. This suggests a nuanced understanding of the risks involved and a recognition of the potential benefits outweighing perceived threats.

The importance of top management support in facilitating technology adoption has been emphasized in the literature (Saeed et al., 2014; Ahlan & Ahmad, 2013). Our results confirm that strong support from top management positively influences cloud adoption initiatives within SMEs. This is consistent with the idea that leadership endorsement fosters organizational commitment and resource allocation towards technology implementation.

Conclusion

Based on the findings discussed above, several conclusions can be drawn regarding the relationship between cloud computing adoption and SME performance: Cloud computing adoption offers significant benefits to SMEs, including increased efficiency, flexibility, and scalability in managing business operations. Leadership support and organizational readiness are critical factors influencing the successful implementation of cloud-based systems within SMEs. Addressing security concerns and leveraging supplier support are essential for building trust and confidence in cloud computing solutions. Competitive pressure acts as a catalyst for driving cloud computing adoption among SMEs, prompting organizations to embrace technology for maintaining competitiveness and achieving growth.

Recommendation

Based on the conclusions drawn from the study, the following recommendations are proposed for practitioners and suggestions for future research: SMEs should prioritize leadership buy-in and organizational commitment towards cloud computing adoption initiatives. Organizations should invest in robust data security measures and risk management strategies to address concerns related to data privacy and security.

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